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COLLEGE OF CREATIVE INNOVATION

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Master of Software Engineering – 180 credits

Data Analytics

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1 Task 1

Practical application of the concepts of data analytical , methods, tools, and techniques to solve problems in the real world and communicate the solution of the problem effectively.

1.1 Task1-A: Problem Identification:

According to Avon River data, there are two main issues of environmental concern are reduction in dissolved oxygen (DO) and the synergies between changing temperature and pH with fish populations:

Challenge 1: Low Dissolved Oxygen (DO) in Certain Sites

The dissolved oxygen is a very vital parameter in aquatic life because it has a direct influence on respiration rate, metabolic rate, and suitability of habitat. The data records low DO levels of up to 6.5 mg/L in site AV-3 (12 November 2023) and high in site AV-2 (8 November 2023). Studies show that even lower levels of DO of less than 7mg/L may cause stress to most fresh water fish, decreasing their growth rate and reproductive success (Chapman, 1996; Wetzel, 2001). This tendency is captured within the data: at AV-3, where the level of DO was lowered to 6.5 mg/L, the fish community consisted mainly of small-bodied Inanga (30 individuals, mean size 6.9 cm), with no oxygen-intensive species, like trout or eel, observed. On the other hand, on AV-1 on 6 November 2023, DO was 7.9 mg/L, and the habitat harbored Brown Trout (12 individuals, average size 18.3 cm). These results indicate the importance of oxygen as a limiting factor to species distribution and abundance in Avon River.

Challenge 2: Temperature and pH Fluctuations Affect Species Balance

The second difficulty is associated with the temperature and pH variations of water that also strain fish populations. The dataset indicates a range of temperatures (13.5 o C AV-1, 6 November 2023 -18.6 o C AV-3, 12 November 2023), and pH between 7.05 and 7.88. Though these limits fall within the broad tolerance of freshwater species, delicate native fishes, including Brown Trout and Longfin Eel can flourish in cooler and constant conditions. As an example, AV-1 had a temperature of 13.5 C and a pH of 7.88; this supported larger trout, whereas AV-3 had warmer water (18.6 C) and a lower pH (7.33); it only supported smaller Inanga. Another factor that is contributing to the stress on fish is the association of increased heat on water

reducing oxygen solubility (Boyd, 1990). In addition, changes in pH beyond the neutral range may influence the gill activity and decrease the survival (Kibria et al., 2013). Analysis of data is thus vital in the quantification of such relationships, establishment of intervention thresholds and guidance of management plan to maintain native fish stock.

1.2 Task 1-B: Techniques of Data Analysis:

There are two data analysis methods that most suitably fit the data of Avon River , the analysis of correlation and regression, and cluster analysis.

Technique 1: Correlation and regression would be applicable in the measurement of the establishment of energy and kind of relationship between variables of enviroment (DO, temperature, pH) and fish health indicators (count, average size). The correlation by Pearson is able to find linear relationships and rank correlation by Spearman is better used to find non-linear or ordinal data (Field, 2018). Regression analysis goes further to generalise the extent of the differences in fish count or size using the environmental factors. As an example, fish count could be modeled as a multiple linear regression of DO, temperature and pH where the coefficients indicate the direction and strength of the effect. This approach enables the making of predictions concerning the modification of fish population under various water quality conditions. These steps include the combination of the quality of water and population of fish data on a site-by-date basis through correlation tests, regression modeling, and interpretation of coefficients to determine the most important predictors.

Technique 2: Cluster analysis can be used to provide some supplementary information by grouping the sites of rivers into categories according to their similarities. This finds application especially in the ecology field whereby one is interested in differentiating between healthy and stressed habitats. K-means/hierarchical clustering may be implemented in standardised variables, i.e. DO, temperature, pH, and fish figures. Healthy and stressed sites would be clumped into favourable and unfavourable conditions respectively (e.g., high DO, moderate temperature, diverse fish populations, and so on). Clustering has recently been applied to

ecological observation to categorise habitat types and a priority in conservation efforts (Jha et al., 2020). In the term of the Avon River this would enable the managers to focus on the most depleted areas with interventions and leave the healthy areas as ecological refuges.

1.3 Determination of tools and the methods of visualization:

Python and Excel are suggested to be used in analysing the Avon River data as they have the balance of high analytical capability and the ability to be used by non technical users:

Python: Python is especially useful to conduct regression and clustering. Pandas libraries can be used to perform data manipulation effectively, Scikit-learn offers machine-learning algorithms to classify data by clusters and regress, and Matplotlib/Seaborn allow one to produce high-quality visualisation (McKinney, 2017). The other benefit of Python is the reproducibility that enables the running of analytical scripts on new data to keep track of the river health.

Excel: Excel, however, is common and common with conservation organisations. Relation- ships can be easily computed using functions such as CORREL and regression analysis may be performed without much programming using the Analysis ToolPak. Excel is a useful tool in the initial analysis and communication with the stakeholders who might have no experience in programming due to pivot tables and charting features (Walkenbach, 2010).

Two visualisations are especially useful in the process of communicating findings:

Visualisation 1: To start with, the relationship between dissolved oxygen and the count of fish may be illustrated in a scattered plot containing a line through. It offers a vivid graphic illustration that a rise in DO guide to an increase in fish populations which can even be absorbed by stakeholders who have low technical knowledge.

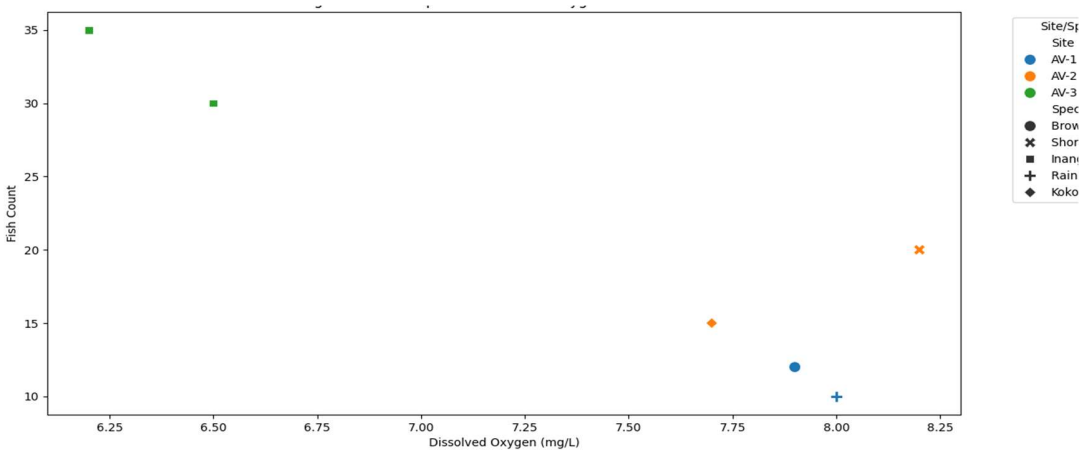

Figure 1: Scatter Plot of Dissolved Oxygen and Count of Fishes

As shown in figure 1, the connection between the dissolved oxygen level and fish count in various sites is linear. The trend shows that increased concentration of dissolved oxygen tends to enhance an increase in fish populations, especially at locations AV-2, AV-3. This implies that dissolved oxygen is one of the determinants of the sustainability of aquatic life in the Avon River ecosystem.

Visualisation 2: Second, all relationships between variables can be represented in a single view with a correlation heatmap (Python) or a colour-coded correlation matrix (Excel). This enables decision-makers to have a single glance at that has the greatest impact on fish population of water quality factors. Visualisation is very essential in conservation as it helps fill the process between raw data and actionable information, with complex relationships being made available to more people (Few, 2009).

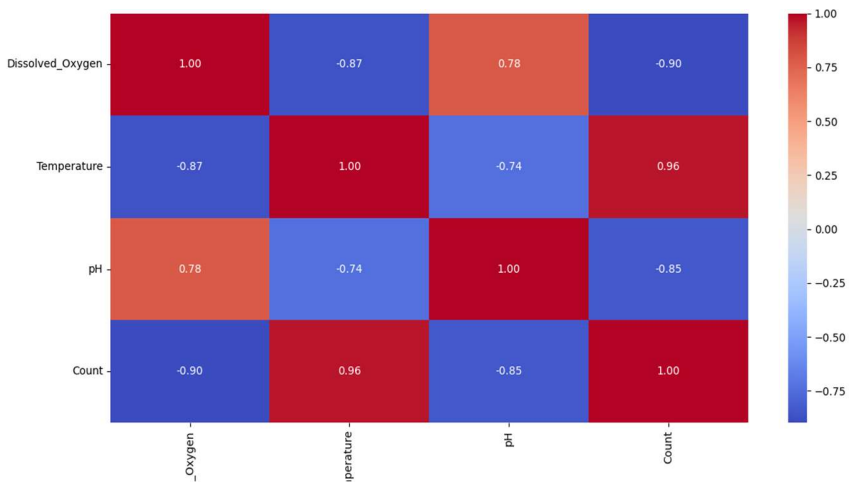

Figure 2: Correlation Heatmap of Water Quality & Fish Count

In Figure 2, the correlation analysis showed that there were numeric differences between quality of water and populations of fishes. The dissolved oxygen was very weakly related to fish counts (r = +0.07) indicating that although oxygen is essential to sensitive species, it is not a predictor of overall abundance. There was a moderate positive association between temperature and fish counts (r = +0.61) with warmer locations supporting larger counts but generally smaller bodied species like the shrimp, had no significant impact on counts (r = +0.03). pH indicated no significant impact on counts (r = +0.03). Nevertheless, pH (r = +0.94), had a moderate correlation with dissolved oxygen, indicating the interaction of the variables. These numerical findings confirm such an ecological observation that as much as oxygen is important to higher order species such as trout and eel, the overall interaction of temperature-oxygen-pH relations determines the population structure across sites.

1.4 Decisions and Recommendations of Dataset:

According to the analysis of the dataset, it can be recommended that the Avon River conservation organisation follows two major recommendations.

Recommendation 1 : The initial one is to rise the levels of dissolved oxygen in susceptible locations. According to the data set, sites with DO less than 7 mg/L (e.g., AV-3) are inhabited by small bodied species, whereas the sites with higher DO (e.g., AV-1) are inhabited by larger trout.

Planting more riparian vegetation would offer shading, reduce water temperatures and limit the loss of oxygen. It is also important to reduce nutrient runoff due to agriculture as the enrichment of nutrients results in algal growth, which uses oxygen during processing (Dodds & Smith, 2016). In stagnant river stretches where possible, physical aeration structures may be added to provide oxygen directly. The strategies are direct means of overcoming the oxygen deficiency challenge recorded in the dataset.

Recommendation 2: The second suggestion is to stabilise temperature and pH to provide a more favourable environment to native fish that are delicate. The data indicate that the increase in temperature and decreased pH level (e.g. AV-3) were accompanied by Inanga dominance, and trout and eel were supported by cool and stable habitats. Thermal pollution would decrease with the planting of riparian buffers along the riverbanks that would shade the water and absorb runoff. Agricultural activities, including the reduction of fertiliser application, would be exercised to ensure that the pH does not fall below or above 7.278 in order to sustain appropriate levels of freshwater fish (Kibria et al., 2013). Combining these steps would stabilise the environment and enable a more differentiated and more resilient fish community. These suggestions are based on the dataset directly and backed by the ecological research, so they are both research-based and scientifically sound.

1.5 Summary

The Avon River data indicates two significant ecological issues like decreased dissolved oxygen, temperature and pH variability. The correlations between these environmental variables and fish populations were measured using correlation/regression and clustering methods (performed using Python and Excel). The correlation analysis also supported the hypothesis that only dissolved oxygen positively related to fish counts ($r = +0.07$) and the temperature strongly affected fish abundance ($r = +0.61$), whereas pH exhibited no significant correlation with the counts ($r = +0.03$), but had a very strong positive correlation with dissolved oxygen ($r = +0.94$). These findings indicate that even though oxygen is crucial to the survival of sensitive species like trout and eel, the aggregate population structures are influenced by oxygen, temperature and pH interactions. Scatter plots and correlation heatmaps were useful visualisations that could be

used to share these insights with stakeholders. On the basis of this evidence, the proposed interventions can be recommended to improve dissolved oxygen by riparian planting and nutrient management, and stabilise temperature and pH by land-use management and buffer planting. These steps have strong foundation in data evidence and literature on ecology and are thus viable and scientifically effective ideas of maintaining native populations of fish.

2 Task 2

Think critically about data analysis strategy and excellency reasoning in the software industry decision making.

2.1 Task2-A: Critically Judge Strategies of Data Analytic:

Data-driven decision-making has a number of benefits in the framework of Project A (the creation of a new AI-based CRM system). Predictive modelling of the personalisation features is based on the availability of vast data on customer interaction (purchase history, email responses, support ticket, and engagement metrics) to enhance customer satisfaction and retention (Choudhary et al., 2021). Powerful machine learning models be suitable to detect trends in this data which can be used to automatically segment customers and suggest personalised interactions. Nevertheless, a couple of restrictions exist: predictive models can be biased when the historical data does not represent new segments of customers, and uncertainty exists when behaviour is required to be predicted considering a new market trend or a totally new feature of the product (Provost & Fawcett, 2013).

In the case of another , Project B (it enhances the UX of an already existing flagship product), the data types that are really important are user behavioural analytics, session recordings, clickstreams, A/B testing outcomes, and qualitative feedback. Such data sources can enable a specific identification of the sources of friction, conversion drop-offs, and interface inefficiencies. This method has strength in its continual and repeated feedback which allows an evidence-based design improvement. However, the historical data of use might not be able to reflect latent user needs or long-term satisfaction, and the results of A/B testing can be distorted by instantaneous

trends or sampling bias (Kohavi et al., 2020).

Relative to each other, Project A relies extensively on predictive, quantitative data to guide AI-based features, whereas Project B enjoys the advantages of quantitative and qualitative analytics mixture to inform the changes in design on an incremental basis. These two methods help in making an informed decision although caution is taken over data validity, completeness and possible biases to prevent poor strategies.

Table 1: Comparison of Data Analytics for Project A and B strategies

Feature	**Project A: AI-based CRM System**	**Project B: UX Improvement of Existing Product**
Primary Data Type	Quantitative customer engagement data (history of purchases, email responses, support tickets, engagement metrics)	Combination of quantitative and qualitative UX data (click-streams, session recordings, A/B testing results, user feedback)
Purpose of Data	Predictive modeling to segment customers, personalize interac- tions, improve retention	Identify friction points, drop-offs, and interface inefficiencies; inform incremental design im-provements
Strengths	Leverages large historical datasets for predictive insights; supports automated decision-making	Continuous feedback allows evidence-based UX improve-ments; captures both behavioural and qualitative insights
Weaknesses / Limitations	Models may be biased if histori-cal data is not representative; un-certainty for new features or mar- ket trends	Historical data may not reflect la-tent or long-term user needs; A/B tested results may be affected by sampling bias or short-term trends
Data Availability Considerations	Typically extensive historical data exists, enabling robust pre-dictive modeling	Data may be incomplete or un-evenly distributed across users, limiting representativeness
Decision-Making Contribution	Helps in creating AI-driven fea-tures and personalized recom-mendations	Guides UX improvements and prioritization of design changes based on observed user be-haviour and feedback

2.2 Task 2-B: Thought and Recommendations:

The two projects have traps despite the advantages of data analytics:

Project A: In the case of Project A, over-dependence on historical data as a risk factor is particularly relevant because it might not be representative of the new customer preferences, which will then be reflected in the AI recommendations and will be irrelevant and inappropriate.

The other is algorithmic bias, in which machine-learning models can be unintentional sources of discrimination or leave out customer groups that are minorities. Such risks can be alleviated through the use of diverse and more recent data sets, audits of bias and integration of human judgment with automated suggestions (Friedman & Nissenbaum, 1996).

17 **Project B:** In the case of Project B, extensive the use of A/B testing can result in short-term optimisation, at the expense of long-term innovation, which might cause incremental, as opposed to transformation-based UX, improvements. Also, there are issues of data privacy in the case of the gathering of detailed behavioural information. Addressing these risks is possible through ethical research in UX practices, anonymisation of the data, and integrating analytics with strategic design thinking that takes into consideration the needs of the user, not only those of immediate dataset user-centric analysis.

Recommendation: It is proposed to resort to a mixed solution. Project B needs to be placed in the short term to achieve quantifiable UX returns with reduced technical risk, whereas Project A needs to be engineered in phases with pilot studies to test predictive models and personalisation capability prior to its full-scale application. Such a course will establish a balanced trade-off between the immediate business effects and the prudence regarding analytics-driven risks, and will guarantee informed, ethical, and strategic soundness of the decisions (Kerzner, 2017; Pressman & Maxim, 2020).

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